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INFOPARK – THE COLLEGE OF IT AND MANAGEMENT

PROJECT REPORT

MACHINE LEARNING

INDIAN STREET FOOD BODY DAMAGE PREDICTOR

Using Linear Regression

Programme / Class	BCA
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Academic Year	2026–2027

Submitted in partial fulfilment of academic requirements.

Project Title: **Indian Street Food Body Damage Predictor Using Linear Regression**

For educational & health awareness purpose only.

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1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. It is used in many fields, including healthcare, finance, transportation and food technology.

This project, **Indian Street Food Body Damage Predictor**, demonstrates how a Machine Learning model can estimate the yearly impact of regular street food habits on the body. The user enters how many times they ate each street food in the last 30 days, and the model predicts an **Annual Body Damage Percentage**.

The project uses Python and Scikit-learn to train a Linear Regression model, and Streamlit to provide an interactive web interface. It covers the basic Machine Learning workflow: dataset preparation, model training, prediction and evaluation.

Note: *This project is made for education and health awareness. The damage percentage is an illustrative score produced by a student model. It is not a medical diagnosis or medical advice.*

2. Problem Statement

People eat street food such as samosa, momo, chowmein and pani puri every day, but most do not know how much long-term damage regular consumption can cause to the body. One plate is not the problem; eating it daily for years is.

There is no simple tool that shows the yearly impact of regular street food habits. A complex model would also be hard for ordinary users to understand. The problem is therefore to build an easy, explainable Machine Learning model that learns the relationship between how often each street food is eaten and the resulting Annual Body Damage Percentage.

The model's performance is measured by comparing predicted values with the actual target values in the test dataset.

3. Objectives

The main objectives of this project are:

- To understand the fundamentals of Machine Learning and Linear Regression.
- To prepare a dataset of street food eating frequencies and body damage values.
- To identify the input features (food frequencies) and the target variable (Annual Body Damage %).
- To divide the dataset into training and testing sets using an 80:20 split.
- To train a Linear Regression model using Scikit-learn.
- To predict Annual Body Damage % from a user's eating habits.
- To evaluate the model using MAE, MSE, RMSE and R^2 score.
- To build a simple Streamlit app that shows the damage percentage, risk level and the biggest culprit food.
- To spread awareness about the effect of oil, refined flour, salt and sugar in everyday street food.

4. Tools and Technologies

Tool	Use in this project
Python	Main programming language used to write and run the Machine Learning program.
Pandas and NumPy	Data handling, inspection and numerical calculations.
Scikit-learn	Splitting the data, training the Linear Regression model and evaluating predictions.
Streamlit	Interactive web interface with sliders, a Calculate button and result display.
Google Colab / Jupyter	Development environment for training and testing the model.
CSV	Storing the dataset in tabular form.

5. Dataset Description

The project uses a dataset of **50,000 samples** covering **10 Indian street foods**. The dataset is designed to demonstrate the relationship between how often a person eats street food and the estimated yearly damage to the body.

The model inputs (features) are the number of times each of these foods was eaten in the last 30 days (value from 0 to 30):

Feature (10 inputs)	Description
Samosa	Times eaten in last 30 days (0–30)
Momo	Times eaten in last 30 days (0–30)
Chowmein	Times eaten in last 30 days (0–30)
Patties	Times eaten in last 30 days (0–30)
Burger	Times eaten in last 30 days (0–30)
Pani Puri	Times eaten in last 30 days (0–30)
Pizza	Times eaten in last 30 days (0–30)
French Fries	Times eaten in last 30 days (0–30)
Soya Bari	Times eaten in last 30 days (0–30)
Dosa	Times eaten in last 30 days (0–30)

Target Variable: Annual Body Damage % — a value from 0% to 100%. A higher value means more damage.

Before model training, the dataset is cleaned: missing values are handled, duplicate rows are checked, and the final feature list is prepared.

Because the dataset is generated for demonstration, it should not be treated as a substitute for real medical or nutritional data.

Why these foods are considered risky

The damage score is based on common harmful ingredients found in street food:

- **Refined flour (maida):** fibre is removed, so it digests fast and spikes blood sugar. Found in samosa, momo and noodles.
- **Reused frying oil:** oil heated again and again forms trans fats that harm the heart. Found in samosa, fries and patties.
- **Too much salt:** excess sodium raises blood pressure over time. Found in chutneys, sauces and chowmein.
- **Added sugar and sauces:** empty calories that cause weight gain and sugar spikes. Found in ketchup and sweet chutney.

In general, such habits are linked to heart risk, weight gain, sugar spikes and high blood pressure. The food itself is not evil; it is oil, maida, salt and sugar in big amounts, every day, that cause the problem.

6. Methodology

The project is carried out in three phases.

Phase 1 – Data Preparation

- **Step 1** — Collect or generate the dataset of 50,000 rows covering 10 foods.
- **Step 2** — Define the features (X): the 30-day frequency of each food.
- **Step 3** — Define the target (y): Annual Body Damage %.
- **Step 4** — Clean missing values and check for duplicates.

Phase 2 – Model Training

- **Step 5** — Split the dataset into training and testing sets using an 80:20 split.
- **Step 6** — Fit the Linear Regression model on the training data.
- **Step 7** — Obtain the model coefficients and intercept.
- **Step 8** — Evaluate the model on the test data using MAE, MSE, RMSE and R^2 score.

Phase 3 – Application

- **Step 9** — Build the Streamlit user interface with 30-day input sliders.
- **Step 10** — Add a Calculate button that triggers the prediction.
- **Step 11** — Display the Damage %, the Risk Level, contribution bars and a tip about the biggest culprit food.

7. Algorithm: Linear Regression

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values. It learns a mathematical relationship between input variables and a target variable.

Since Annual Body Damage % is a continuous value and the pattern between food frequency and damage is treated as linear, Linear Regression is a suitable choice. It is also fully explainable: every coefficient directly shows how harmful the model considers a food to be.

The general multiple Linear Regression equation for this project is:

Damage % = Intercept + Sum of (Coefficient × Frequency of each food)

That is: Damage % = Intercept + (b1 × Samosa) + (b2 × Momo) + (b3 × Chowmein) + … + (b10 × Dosa)

The model learns the coefficients and intercept from the training dataset. After training, it uses the supplied frequencies to estimate the yearly damage. The quality of the predictions is evaluated on the test dataset.

Key coefficients

A higher coefficient means that eating the food one more time in 30 days adds more to the predicted damage.

Food	Coefficient
Samosa	0.755
Momo	0.736
Chowmein	0.694
Patties	0.674
Burger	0.604
Pani Puri	0.475

Samosa has the highest coefficient among the foods listed, followed by Momo and Chowmein, while Pani Puri has the lowest. Samosa and momo are made of maida and are deep-fried or served with salty, oily chutney. Chowmein carries a lot of oil and sauce sodium with little fibre. Pani Puri has a fried shell and sugary chutney, and unclean water can also cause stomach infection.

8. Results and Evaluation

The model was evaluated on the 20% test data, which it never saw during training. The following metrics were obtained:

Evaluation Metric	Result
Mean Absolute Error (MAE)	~1.6 (damage % points)
Mean Squared Error (MSE)	_____ (copy from notebook)
Root Mean Squared Error (RMSE)	_____ (copy from notebook)
R^2 Score	~0.96

The R^2 score of about 0.96 indicates that the model explains approximately 96% of the variation in the damage percentage on the test data. An MAE of about 1.6 means that, on average, the predicted damage differs from the actual value by about 1.6 percentage points.

These results indicate a strong fit to the project's dataset. They do not establish equivalent accuracy for real people or real health outcomes.

9. Example Prediction

Once the model is trained, a user provides the 30-day frequency for each food: Samosa, Momo, Chowmein, Patties, Burger, Pani Puri, Pizza, French Fries, Soya Bari and Dosa.

The model processes these inputs and returns the Annual Body Damage %, a Risk Level and the food that contributes the most.

How to read the coefficients: if a person eats samosa 10 times in 30 days, it adds 10 × 0.755 = 7.55 percentage points to the predicted damage, with all other foods kept the same. The same 10 plates of pani puri would add 10 × 0.475 = 4.75 points.

Example workflow:

1. Enter the number of times each food was eaten in the last 30 days.
2. Pass the values to the trained model in the correct feature order.
3. Generate the predicted Annual Body Damage %.
4. Display the damage %, risk level, contribution bars and tip.

Note: The exact full example output (final damage % and risk level) should be copied from the executed notebook or app so that the report matches the actual implementation.

10. Model Evaluation Notes

Mean Absolute Error (MAE): The average of |actual - predicted|. It is the average gap between actual and predicted damage %. Lower is better.

Mean Squared Error (MSE): The average of (actual - predicted) squared. Squaring means big mistakes are punished more. Lower is better.

Root Mean Squared Error (RMSE): The square root of MSE. It comes back to the same units as the target, which is damage %. Lower is better.

R^2 Score (Coefficient of Determination): R^2 = 1 - (model error ÷ total variation). It shows how much of the damage % the model explains. A value of 1 is perfect, so closer to 1 is better.

All metrics should be considered together. A strong result on the project dataset does not guarantee reliable performance in real life.

11. Application (Streamlit App)

The trained model is wrapped in a simple Streamlit web app so that anyone can use it. The user flow has four steps:

Step	Action	What happens
1	Open App	The Streamlit web app opens in the browser.
2	Set Sliders	The user sets how many times each food was eaten in the last 30 days.
3	Click Calculate	A button at the bottom triggers the prediction.
4	See Results	Damage %, Risk Level, contribution bars for each food and a tip about the biggest culprit food.

12. Advantages

- Demonstrates the complete basic Machine Learning workflow.
- Uses a simple and fully explainable algorithm; coefficients show which food is more harmful.
- Shows how multiple input features can be combined to make one prediction.
- Provides numerical metrics (MAE, MSE, RMSE, R^2) to judge model performance.
- Has an easy interactive interface that anyone can use.
- Spreads awareness about everyday food habits.

13. Limitations

- The dataset is generated for demonstration and is not based on clinical or medical studies.
- Real body damage depends on many other factors such as age, exercise, genetics, portion size and overall diet, which are not in the model.
- Linear Regression assumes a linear relationship and may not capture complex, nonlinear effects.
- The damage percentage is only an illustrative score and should not be used for medical decisions.
- Only 10 street foods are included.

14. Future Scope

The project can be improved in the following ways:

- Use real nutritional and health data, with expert guidance, for training and testing.
- Add more street foods and portion size as an input.
- Add personal factors such as age, weight and activity level.
- Compare Linear Regression with other regression algorithms.
- Give healthier alternative suggestions for each food.
- Deploy the app online and make it available on mobile.

15. Conclusion

The Indian Street Food Body Damage Predictor demonstrates how Machine Learning can estimate a continuous value, the Annual Body Damage %, from multiple input features. The project covers dataset preparation, feature selection, training and testing, prediction, evaluation and a working Streamlit application.

The model achieved an R^2 score of about 0.96 and an MAE of about 1.6 on the 20% test data. The coefficients show that foods such as samosa, momo and chowmein carry the highest weights, while pani puri carries the lowest among the foods analysed.

The project gives a practical introduction to regression and a simple message: one plate is not the problem, but eating it daily for years is. Further work with real health data would be needed before the tool can be used beyond education and awareness.

16. References

1. Scikit-learn documentation — Linear Regression and model evaluation.
2. Pandas documentation — Data handling and analysis.

3. NumPy documentation — Numerical computing.
4. Streamlit documentation — Building interactive web apps in Python.
5. Python documentation — Programming language reference.
6. Project dataset: Dataset of 50,000 samples created for this demonstration.
7. Project notebook and app: Indian Street Food Body Damage Predictor.

17. Declaration

I hereby declare that the project report titled “Indian Street Food Body Damage Predictor using Linear Regression” has been prepared as part of my academic work for the Machine Learning subject.

The report describes the project methodology, implementation and evaluation. The results are presented as obtained from the project’s test evaluation, and the project is meant only for educational and health awareness purposes.

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